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## **EC-5005(1)-CBGS**

### **B.E. V Semester**

Examination, December 2020

### **Choice Based Grading System (CBGS)**

### **Computer System Organization**

*Time : Three Hours*

*Maximum Marks : 70*

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Answer the following questions:
  - i) What is register transfer language? Clear it with example.
  - ii) Explain the operation of three state bus buffers and show its use in design of common bus.
- b) Discuss the phases of Instruction Cycle with flowchart.
2. a) Draw the circuit for control unit of basic computer and explain its working.
- b) What is the difference between microprocessor and micro program? Is it possible to design a microprocessor without a micro program? Are all micro programmed computers also microprocessors? Define micro operation, micro instruction, micro program and micro code in short.
3. a) Explain the process of Addition, Subtraction and multiplication in ALU.
- b) What are the different modes of data transfer between I/O devices and memory ? Explain any one of them in detail.

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4. a) Differentiate simplex, half duplex and full duplex data transmission.  
b) Draw block diagram and explain Daisy chaining method of establishing priority.
5. a) Write short notes on virtual memory organization. What do you mean by Paging?  
b) What are the three memory mapping procedures of cache memory? Explain using suitable examples.
6. a) Explain how address sequencing of control memory is achieved in a micro programmed control unit?  
b) What is microinstruction format? Explain different field of microinstruction.
7. a) Is the advantage of pipelining. Discuss pipeline hazards in detail. Differentiate instruction and arithmetic pipelines in short.  
b) A non pipeline system takes 50ns to process a task. The same task can be processed in a six segment pipeline with a clock cycle of 10ns. Determine the speedup ratio of the pipeline for 100 tasks. What is the maximum speedup that can be achieved?
8. a) Give a summary of arithmetic and logical operations that are defined for the vector architecture?  
b) Explain how pipelining speed ups the processing of an operations? Explain the working of a four segment instruction pipeline.

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